

BNY MELLON – INTERNSHIP INTERVIEW EXPERIENCE

- **Ajai Krishna T.S.**
2021103503
CSE Batch of 2025, CEG

About the company

The Bank of New York Mellon Corporation, is an American investment banking services holding company headquartered in New York City. BNY Mellon was formed from the merger of The Bank of New York and the Mellon Financial Corporation in 2007. It is the world's largest custodian bank and securities services company. The bank's primary functions are managing and servicing the investments of institutions and high-net-worth individuals. Its two primary businesses are Investment Services and Investment Management. The bank's clients include 80 percent of Fortune 500 companies. The Mumbai Representative Office of The Bank of New York Mellon was set up in 1983 and continues to serve as BNY Mellon's main liaison point for Indian client base. In 2013, Pershing India Operational Services Private Limited (Pershing India) was established in Chennai. Currently, the company have over 12,000 employees across our three locations: Mumbai, Pune and Chennai

No of Rounds: 3 (Online Test + Technical interview + Technical & HR interview)

Stipend: 75,000 INR per month

Duration of internship: 7 weeks

Round 1 – Online Assessment

The online assessment was conducted on the **Hackerrank platform**, there were a total of 4 questions that were asked, which consisted of 1 easy, 2 medium and 1 hard question

Duration: 2 hours

The questions that I had gotten are as follows:

1. **Category: Medium**
Topic: Graphs

A server network is represented as a tree of `g_nodes` servers indexed from 1 to `g_nodes` and `g_nodes-1` edges where `ith` edges connect the servers `g_from[i]` and `g_to[i]` the transfer time between any 2 connected servers is 1 unit. Given the graph `g` find maximum time taken to transfer the data between any 2 servers in the system.

Write a function `getMaxTime` in python with the parameters:

`int g_nodes`: the no of servers in the network

`int g_from[n]`: the end point of edges

`int g_to[n]`: the end point of edges

Function returns:

`int`: the maximum time to transfer data between 2 servers

2. **Category: Hard**

Topic: Graphs

A game contains a maze represented as a graph with n vertices numbered 1 to n and m bidirectional edges numbered 1 to m . The i th edge connects $\text{endpoint1}[i]$ and $\text{endpoint2}[i]$. It takes $\text{edheLength}[i]$ units of time to traverse the i th edge. Each vertex l will disappear at time $t[l]$ and will not reappear. If a visit occurs at the same time the vertex disappears, the vertex cannot be reached. The game starts at vertex 1 and time 0. Find the earliest visit time for each vertex starting from vertex 1. If a vertex cannot be reached the arrival time is -1 Write python function `mazeGame` that has following parameters

Int $t[n]$: array of integers the i th vertex goes away at $t[i]$ 1 based index

int $\text{endpoint1}[m]$: vertex no of 1 end of an edge $[i]$ 1 based index

int $\text{endpoint2}[m]$: vertex number of the other end edge $[i]$ 1 based index

Int $\text{edheLength}[m]$: $\text{edheLength}[i]$ is the time it takes to traverse edge $[i]$ 1 based index

Returns:

int $[m]$: the arrival times at each vertex l or -1 if visit is not possible

3. **Category: Easy**

Topic: Single Dimensional Arrays

There are n devices in the network where the vulnerability of i th device is represented by $\text{vulnerability}[i]$. All devices in a security level must have the same vulnerability level and the number of devices in any 2 security levels should not differ by more than 1 Given the array `vulnerability`, find the maximum no of security levels needed to ensure protection of network Write a function `getMinimumGroups` that has the parameter:

Int $\text{vulnerability}[i]$: the vulnerabilities of devices

Returns

Int: the minimum no of groups required

4. **Category: Medium**

Topic: Multi-Dimensional Arrays

There are n customers and m shops in HackerMall. The no of items of each type in a shop is unlimited so several customers can buy items of same type in a shop A customer is considered as served if all types of items the customer wishes to

purchase are available in open shops. Initially the shops are closed. Find maximum no of customers who can be served by the mall if almost 2 shops are opened to serve the customers Write function countmaximumcustomers has the following parameters:

int customer[m][p] : the types of items each customer wishes to purchase

int shop[m][k]: the types of items offered by each shop

Returns

Int: the maximum no of customers that can be served

Round 2 – Technical Interview

The interview was conducted in person at CUIIC. It went for about 35-40 min.

The interviewer first introduced herself and in turn asked for my introduction. Then she went through my resume and asked me which programming language I'm proficient in. I told Python, since I was very familiar with it.

She then asked me why I chose Python over the other languages, and I told her its because its an interpreted language and its easy and user-friendly nature and syntax.

The interviewer started to test me in Data Structures. She gave me a piece of paper and asked me to write down each data structure that I have learnt, draw its representation and write functions to perform basic operations like creation, insertion, deletion and search. I started off with arrays, how to access elements in an array and wrote functions for performing the elementary operations. I explained every line of the code as I was writing. I did this for every data structure upto graphs.

While doing this task, I was asked many questions like:

1. What is the advantage of linked list over other linear data structures?
2. Given a binary tree, I was asked to write its inorder, preorder and postorder traversal, and given two out of three traversals, I was used to construct the corresponding binary tree
4. What is the difference between tree and graph
5. Given a graph, write its BFS and DFS traversals and explain the procedure to perform the same

After this was done, the interviewer moved on to asking me basic questions in Python programming, to test my knowledge. I was then asked to write an optimal solution to remove duplicates in a string, and to perform basic arithmetic operations on matrices. I wrote down the programs on paper and explained my approach to her step by step. She was satisfied with it.

Next, the interviewer tested me in DBMS. She gave me a Student database with 4-5 tables in it, and asked me to write SQL queries to retrieve appropriate records from the table. This task involved the use of joins – equijoin, left join, right join, inner join, natural join, triggers and cursors. Given an equijoin query, I was asked to translate it to natural join, left join, right join and natural join. This extensively tested my understanding of join operations and the basic syntax of various types of joins in SQL. It was a bit challenging but interesting.

This was followed by a few general questions in SQL:

1. Difference between order by and group by
2. When do we use having clause
3. Aggregate functions
4. Difference between drop and truncate command

The interviewer concluded by asking me what my favourite subject was, in Computer Science, and I said DBMS. She then asked me if I have any questions for her, and I asked her what tech stacks we'll be generally using for doing projects during the summer internship, and what prior knowledge we are expected to have.

Overall, the interview went well and I was shortlisted for the next round

Round 3 – Technical + HR Interview

It took place in person at CUIIC. It went for about 45 min. I was asked to bring my laptop along with resume.

The interviewer greeted me and asked me to introduce myself. Then he went through my resume thoroughly.

Since I'm doing the BS Data Science and Applications program concurrently with my college degree and I had mentioned it in my resume, the interviewer was curious to know more about the program and its modalities, so I spent a lot of time explaining how I got into the program, why I wanted to pursue it and how I had benefited out of the program overall. He then asked me a few questions he had about the IITM program and I answered them.

After this, the interviewer gave me three questions, I was asked to open my laptop, use any editor of my choice to type out the code and run the programs in front of him. The questions that were given to me were:

1. Given some numbers in an array, use bubble sort and merge sort to sort the numbers in the array, and show the results obtained after every pass
2. Write code to perform division of 2 numbers without using the '/' operator, and multiplication of 2 numbers without using '*' operator
3. Write optimal solution to reverse a number and string, and to check whether it is a palindrome or not

I had to code in the presence of the interviewer and explain every part of the code as I typed. I was asked to write separate functions to perform each operation in a program for modularity.

Following this, I was questioned extensively about the projects that I had put on my resume. I had a music management system project and hotel booking system project, and I was asked to describe what tech stacks I had used for frontend and backend development. I was also told to describe the structure of the database created and how the tables were linked, what was the choice of primary key, foreign key and why. I was also asked which normal form the tables existed in, and how to normalise it completely. The interviewer was also curious to know the unique selling point of my project and what added benefits it offers compared to traditional systems

After this was done, the interviewer looked into my resume and asked me about my positions of responsibilities. Since I was a Deputy Director in CSAU, ACM-CEG, Guindy Times, I explained about my role in each of these clubs and told him what kind of sessions and programs each of these clubs conduct in college. He was eager to know more about these clubs and also suggested a few sessions and events that technical clubs in our college could conduct, to attract many people. It was a very friendly conversation overall.

The interviewer also asked me how I was able to manage being in so many clubs while pursuing a dual degree, and I told him that managing time wisely was the key.

Finally, the interviewer asked me whether I have any questions for him. I asked him about the various teams for summer internship at BNY and what each team does. I then thanked the interviewer and came out.

After an hour, the results were announced.

Overall, 9 of us from CEG and MIT got selected for the summer internship program at BNY Mellon, and I was one among them.

Tips:

1. Be confident with what you put on your resume. Do not fake anything there as you can be questioned on anything there, and you are expected to be thorough with what is mentioned there.
2. Have 2-3 solid projects to put on the resume. This is very crucial, as you are interviewing for a technical role. So, the interviewer would be more interested to check out your current knowledge and gauge your understanding of various tech stacks, to know whether you'll be able to handle any project given to you during the internship
3. Core concepts that you need to know – **DSA,DBMS, OS, OOPS**. This forms the main part of testing in any interview.
4. Never panic if you don't answer a couple of questions in the interview. The interviewer is just seeing how you would act in different situations.
5. Prepare well beforehand for **Tell me about yourself**. Remember - what you say in this is going to drive the rest of your interview, so ensure that you highlight all your strengths, hobbies and achievements in your introduction.
6. Maintain a good rapport with the interviewer. That is very crucial. Keep a smile on your face throughout. Show them that you are genuinely interested about the role.

If you have any further questions, feel free to reach out to me

Ajai Krishna T.S

Batch: 2021-2025

Department: CSE

Contact No: +91 70107 07652

Linkedin Id: <https://www.linkedin.com/in/ajai-krishna-t-s-4b3508243/>

Email id: ajaikrishnats567@gmail.com

All the very best!